



Final Project Data Visualization

Programming Foundation for Digital Media

Presented By

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Table of Contents



Executive Summary	I.
Introduction and Motivation	II.
Data Collection and Methodology	III.
Visual Design (Visual Encodings)	IV.
Alternate Sketch Analysis	V.
Technical Implementation	VI.
Analysis of Results and Findings	VII.
Conclusions and Future Work	VIII.
References	IX

1. Executive Summary

When I started this project, I was motivated by the prospect of seeing the final results. However, as time went on, collecting the data became increasingly difficult, but I was determined to achieve the ultimate goal. I wanted to understand if my physical exertion affects my daily energy levels and, if so, how.

The project combines my own manual energy log with Apple Watch data to create an interactive visualization in D3.js. In this report, I explain how I collected and cleaned the data, how I arrived at the final scatterplot design, why I discarded other ideas, and what conclusions I was able to draw from the patterns that emerged.

2. Introduction and Motivation

Context

This project serves as the culmination of the course, in which we applied what we learned in programming to a more real-world scenario. The idea wasn't just to "create a graph," but to demonstrate that I can organize data and use libraries like D3.js to transform a set of numbers into a useful visualization.

Project Objectives

My main goal was to see how my daily energy levels change depending on the physical exertion I undertake. Instead of simply graphing calories, I wanted to connect several variables: Active Kilocalories, Duration, Heart Rate, and my energy levels before and after exercise. This way, I could see if the effort I exert is actually reflected in my daily energy levels.

Hypothesis

My analysis was driven by the following assumptions regarding physical exertion and fatigue:

- Primary Hypothesis: "As Active Kilocalories increase on any given day, the energy change will more frequently show a negative result."
- Rationale: The idea behind this was that the body has a limited amount of energy and that, if the exercise is very intense, a drop in energy levels should be noticeable. I also thought that longer sessions or sessions with a higher heart rate might amplify this effect.
- Secondary Factor: In addition, I hypothesized that longer workout Duration and high Maximum Heart Rate might amplify the overall effort, leading to a higher probability of Energy Loss.

3. Data Collection and Methodology

Collection Rationale

To ensure a good dataset spanning at least one month, I employed a two method data collection strategy. This approach captured both objective training intensity and subjective physiological response.

- **Objective Data:** I utilized an **Apple Watch** to record workout sessions. This provided consistent measurements for:
 - Active Kilocalories (kcal): Total work output.
 - Duration (minutes): Temporal length of the workout.
 - Maximum Heart Rate (BPM): Peak cardiovascular intensity.
- **Subjective Data:** I maintained a manual log using the "Notes" application. To standardize this data, I created a 5-point scale ranging from 1 (Exhausted) to 5 (Fully Energized). Recordings were made at three intervals: Morning, Post Workout, and End of Day.

🌐 Sanchez_Gabriela_data

Objective Data					Subjective Data*		
Date	Workout Star Time	Duration (min)	Active Kilocalories	Heart Rate (BPM)	Energy level morning	Energy level after workout	Energy level end of day
29 Sep 2025	10:00 am	107	378	74-165	2	4	3
30 Sep 2025	10:01 am	93	313	76-162	3	2	4
1 Oct 2025	10:05 am	108	352	84-165	2	2	3
2 Oct 2025	10:19 am	105	312	76-156	3	1	4
3 Oct 2025	9:57 am	86	386	84-175	1	3	2
6 Oct 2025	10:10 am	106	381	74-168	1	2	2
7 Oct 2025	10:08 am	96	285	73-144	1	3	2
8 Oct 2025	10:17 am	118	465	94-175	1	2	3

*Scale Parameters : 1-Exhausted, 2-Tired, 3-Ok, 4-Energetic, 5-Fully Energized.

Colors Parameters: Went up, Went down, Stayed the same

Screenshot Data Google doc data Preview

Data structure

- **X-Axis:** Date/Day (Temporal/Sequential)
- **Y-Axis:** Active Kilocalories (Quantitative - Intensity)
- **Color:** Max Heart Rate in BPM (Low to High Intensity)
- **Shape (Glyph):** Energy Change (Morning workout vs Post workout)
 - Energy Gain (Triangle ▲).
 - Energy Loss (Triangle ▼).
 - No Change in Energy (Circle ●).
- **Size:** Duration in min (Quantitative)

Data Cleaning and Formatting

The raw data was initially unstructured. To make it compatible with the D3.js library, the following processing steps were taken:

1. **Normalization:** Dates were standardized to ISO 8601 format (YYYY-MM-DD).
2. **Derivation:** I calculated a derived metric: **Energy Change** (Energy Post-Workout minus **Energy Morning**). This was crucial for the visualization, as it allows me to classify the results as "Gain", "Loss" or "No Change".
3. **JSON Export:** The final dataset was structured as an array of objects in **workout-data.json**.

4. Visual Design (Visual Encoding)

Visualization Choice: Time-Series Scatterplot

After trying different ideas, I chose a scatterplot with a timeline because it's clear, practical, and works very well for visualizing relationships between variables. I also found it more intuitive than other options that became overly complicated with so many dimensions.

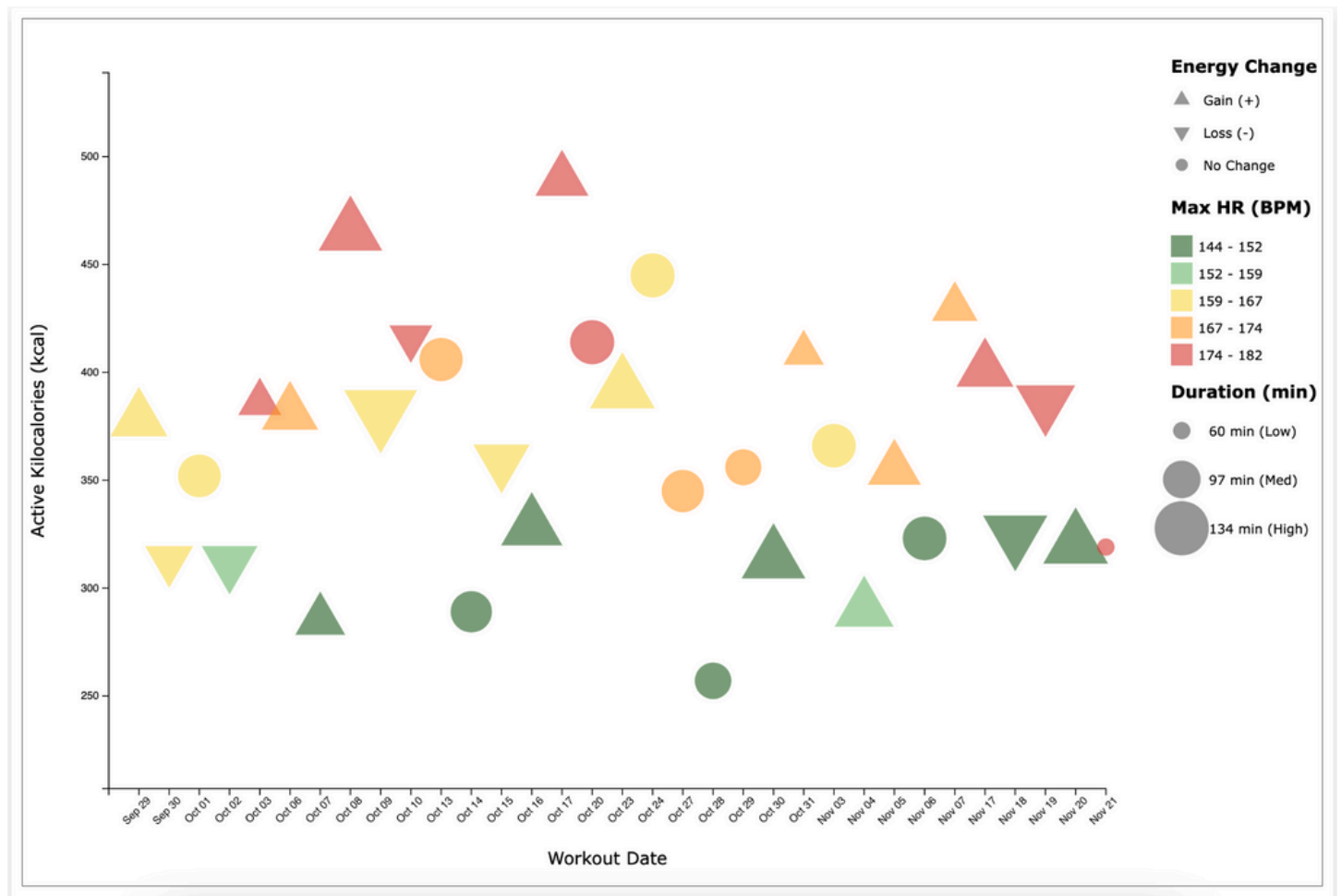
Visual Encodings (Marks and Channels)

To represent the five dimensions I utilized the following encodings:

- **Position (X/Y):** Date vs. Active Kilocalories. This reveals sequential patterns and intensity outliers.
- **Color (Hue):** Max Heart Rate. I used a Quantize scale (Red to Green) to differentiate between low-intensity and high-intensity cardiovascular sessions.
- **Size (Area):** Duration. I used a **d3.scaleSqrt** for the radius. This adheres to visualization best practices, as human perception aligns with area rather than radius.
- **Shape (Glyph):** Energy Change. This utilizes the concept of "Mapping Data onto Aesthetics."
 - **Triangle Up:** Energy Gain (+).
 - **Triangle Down:** Energy Loss (-).
 - **Circle:** No Change.

Scatterplot Description:

Time series scatter plot with date on the X-axis and active kilocalories on the Y-axis. The size of each point represents the duration of the workout, and its color represents the maximum heart rate. The shape is used to show the net change in energy: a triangle if energy increased, an inverted triangle if it decreased, and a circle if there was no change. This allows you to see patterns without performing calculations, simply by interpreting the symbols.



Screenshot of Final Visualization. *Final visualization showing the relationship between workout intensity and energy change.*

5. Alternate Sketch Analysis

During the prototyping phase, I developed a strong candidate titled "**Proposal 3: The Energy Journey.**"

Description

This was a **Parallel Coordinates Plot** featuring four vertical axes: Day of Week, Morning Energy, Post Workout Energy, and End of Day Energy. Each workout was a polyline connecting these axes, with thickness representing Max Heart Rate and color representing Calories.

Critique and Rejection

- **Pros:** It visualized the process rather than just the result, answering questions like "Does a high-energy morning guarantee a high energy evening?"
- **Cons:** With two months of data, the plot suffered from the "hairball effect" (visual clutter). Furthermore, comparing the "Intensity" (color) to the "Energy Drop" (slope) was cognitively difficult. The Scatterplot offered a clearer summary of the *outcome*, which was the priority for this study .

The Energy Journey (Flow)

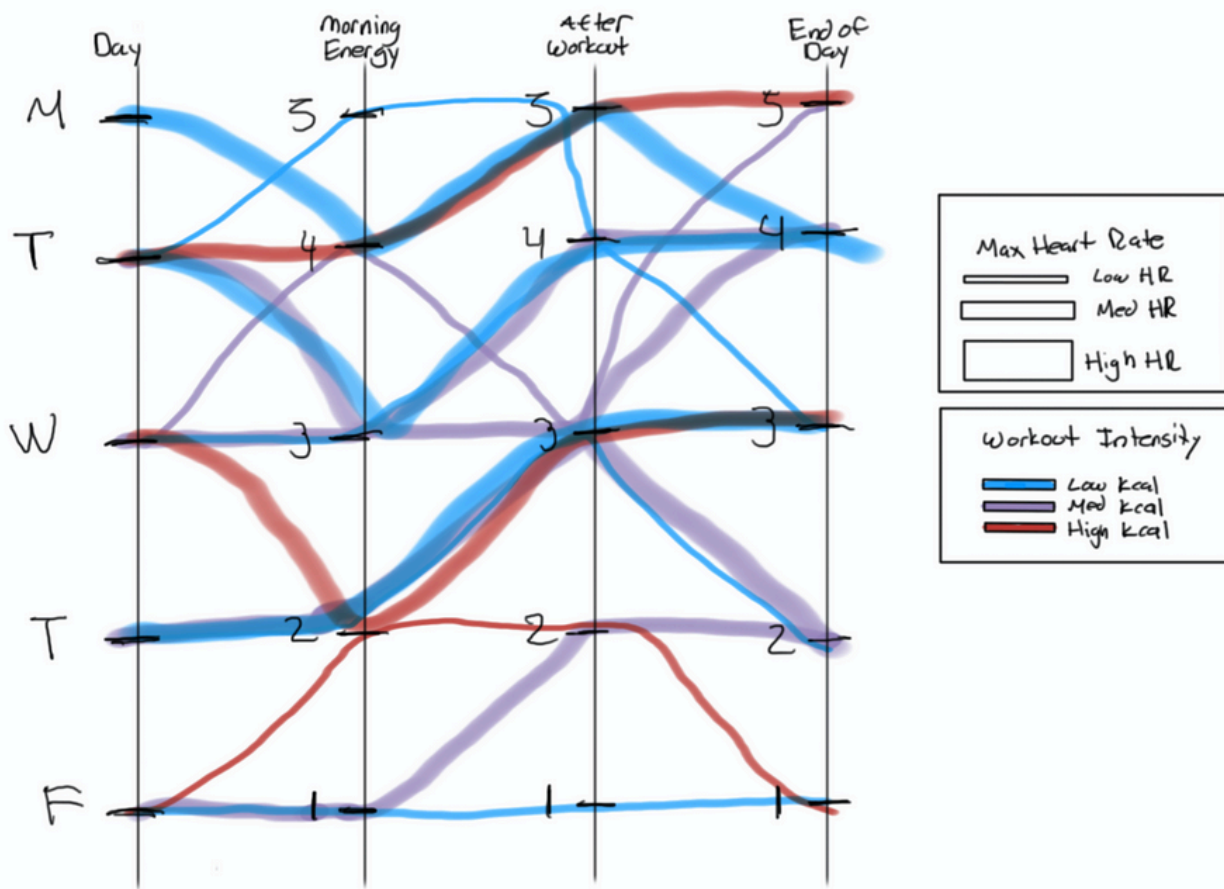


Image of Sketch 3 "Energy Journey"]. Sketch of the rejected Parallel Coordinates Plot.

6. Technical Implementation

Code Structure

The script is divided into three main functions to ensure maintainability:

- **organizeData():** Handles business logic and cleaning.
- **buildScales():** Defines the mathematical relationship between data and pixels.
- **drawVisualization():** Manipulates the DOM and generates SVG elements.

The Mixed Geometry Challenge

One of the biggest challenges was mixing triangles and circles within the same graphic. The solution was to use `d3.symbol()`, which greatly simplified the process. For the inverted triangles, I only used a rotation, without doing any complicated calculations.

Dynamic Legends

To ensure the visualization wouldn't break with future changes, the legends weren't hardcoded with fixed values. Instead, I used `invertExtent()` to get the ranges directly

from the color scale. This makes the visualization automatically update if the data changes or is updated.

7. Analysis of Results and Findings

After the creation of the data visualization, a visual analysis of the emerging patterns was performed.

Refutation of the Intensity Hypothesis

Contrary to my initial expectations, there is no clear clustering of inverted triangles (energy loss) at the top of the Y-axis (high calorie expenditure) or in the dark red areas (high heart rate). The gain and loss glyphs appear unevenly distributed across the intensity spectrum. Key finding: This taught me that exercise intensity alone doesn't tell us much about how my energy levels change.

The "Sweet Spot"

A subtle trend was observed in the middle of the graph. Workouts of medium duration (medium circles) and moderate heart rate (orange/medium shades) showed a slightly higher pattern of ascending triangles (gain).

8 Conclusions and Future Work

Multifactorial Conclusion

The scatterplot suggests that the dependent variable of energy change is influenced by external factors not included in this visualization. It is very likely that variables such as the quality of sleep the night before, nutrition before and after training, and stress levels have a greater impact than the intensity of my exercise.

Future Data Collection

In the future, I'll expand my data collection to include sleep quality and nutrition. It's likely that the days my energy levels dropped were due to lack of sleep or a poor night's sleep, not training intensity. I believe adding that data would provide a much more complete picture of why my energy levels fluctuate on certain days.

9. References

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